Order Selection and Forecasting in Long-Memory Functional Time Series

A unified FARFIMA framework

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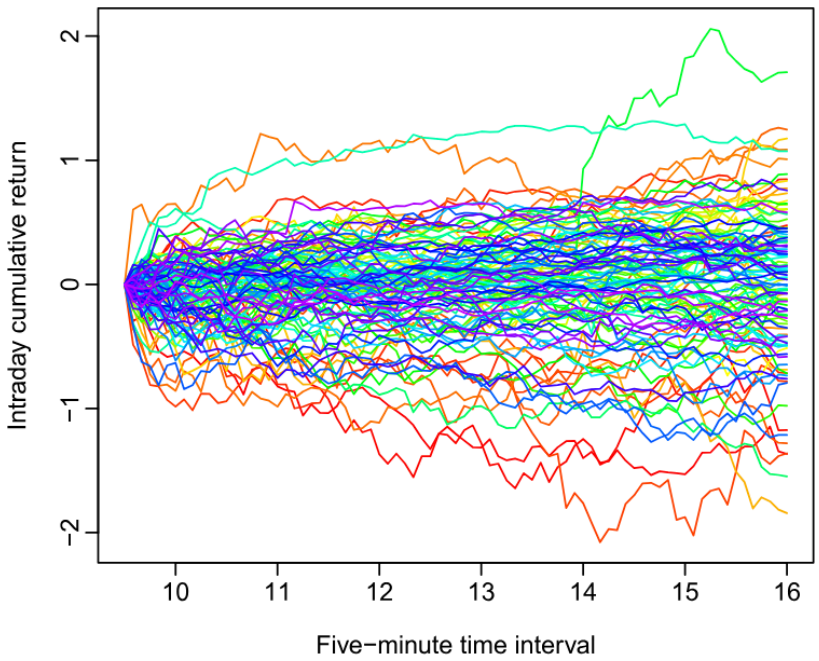
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Outline

1. Introduction
2. Literature & gap
3. Methodology
4. Simulation
5. Conclusion

A functional time series is a sequence of curves



Functional cumulative intraday returns: one curve per trading day.

$$X_t(u)$$

t: which day

u: position within that day

$X_t(u)$: the complete daily shape

Other examples

Electricity-demand curves

Financial yield curves

Highway traffic-flow curves

Example adapted from Shang (2017), *Journal of Forecasting*.

FARMA describes two ways the past enters today

For a stationary functional time series, write the curve at time t as Y_t.

$$Y_t = \underbrace{\sum_{i=1}^{p} \Phi_i Y_{t-i}}_{\text{past curves}} + \eta_t + \underbrace{\sum_{j=1}^{q} \Theta_j \eta_{t-j}}_{\text{past shocks}}$$

Functional autoregression (FAR)

Yesterday's curves directly shape today's curve.

p = number of curve lags

Functional moving average (FMA)

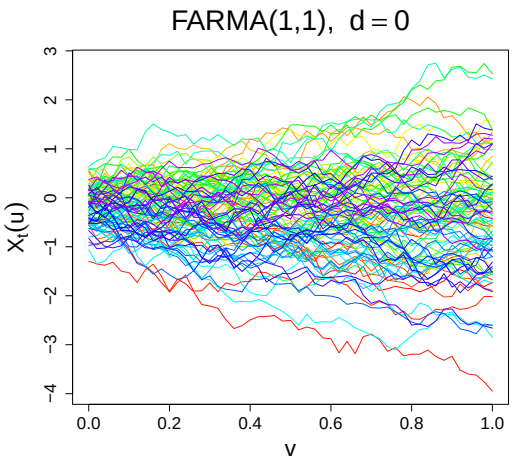
Recent unobserved shocks continue to echo.

q = number of shock lags

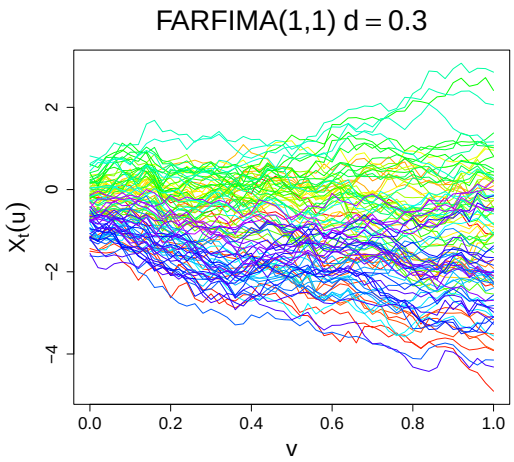
$\Phi_i, \Theta_j : \mathcal{H} \to \mathcal{H}$ map an entire input curve to an entire output curve; $\mathcal{H} = L^2(\mathcal{U})$.

Long memory keeps old shocks alive

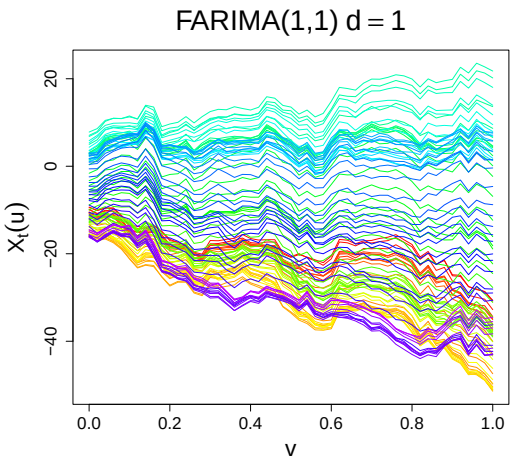
Short memory: $d = 0$



Long memory: $d = 0.3$



Non-stationary: $d = 1$



For $0 < d < \frac{1}{2}$, dependence decays slowly; for $d > \frac{1}{2}$, variability can accumulate and the original process is non-stationary.

FARFIMA(p, d, q) separates three sources of dependence

FARFIMA(p , d , q)

Autoregression

p: number of past-curve lags

Memory

d: persistence parameter

Moving average

q: number of past-shock lags

$$Y_t \sim \text{FARMA}(p, q) \xrightarrow{(1-B)^{-d}} X_t \sim \text{FARFIMA}(p, d, q)$$

Selection problem

Forecasting problem

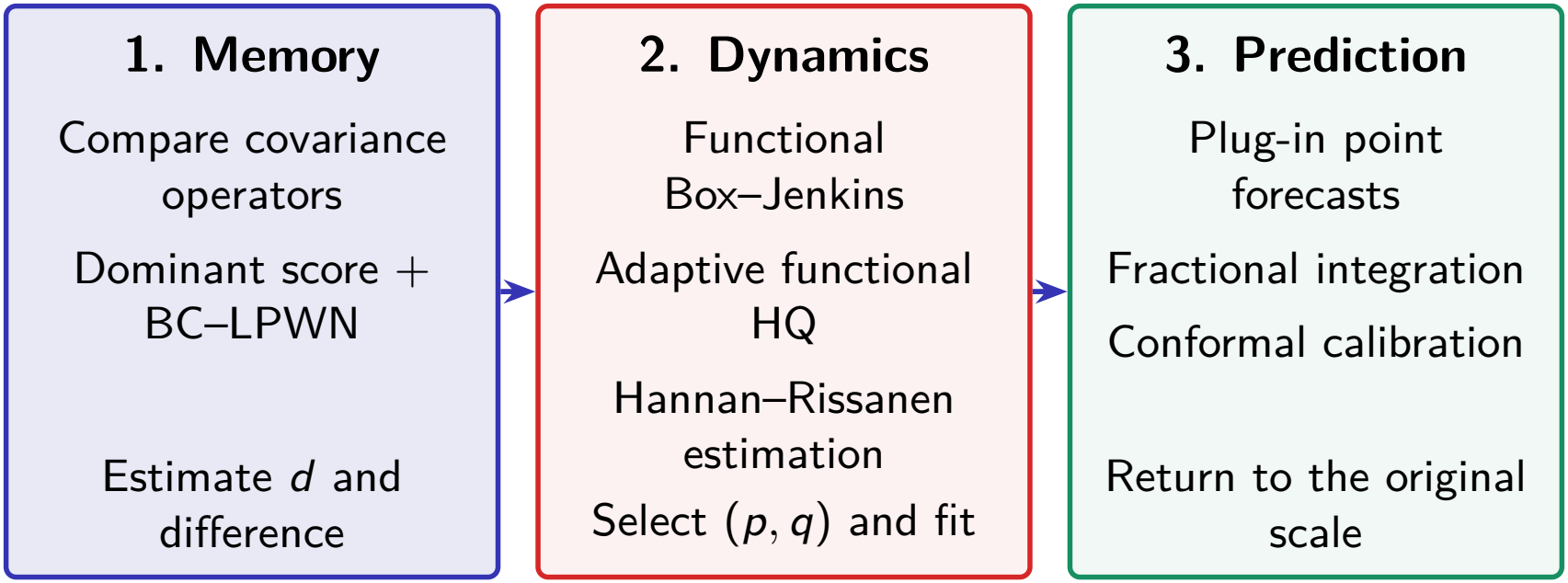
The literature leaves the full workflow disconnected

Stream	What is established	The gap
Estimation of memory parameter	Dominant subspaces and estimation of d	
Functional order selection	fACF/fPACF; sequential FAR tests	Diagnostic or pure-FAR restrictions
Functional forecasting	FARFIMA$(1,1,0)$	No fractional differencing; no moving-average term

Gap: estimate d, select both p and q, estimate the operators, and produce calibrated forecasts on the original functional scale in one theoretically grounded workflow.

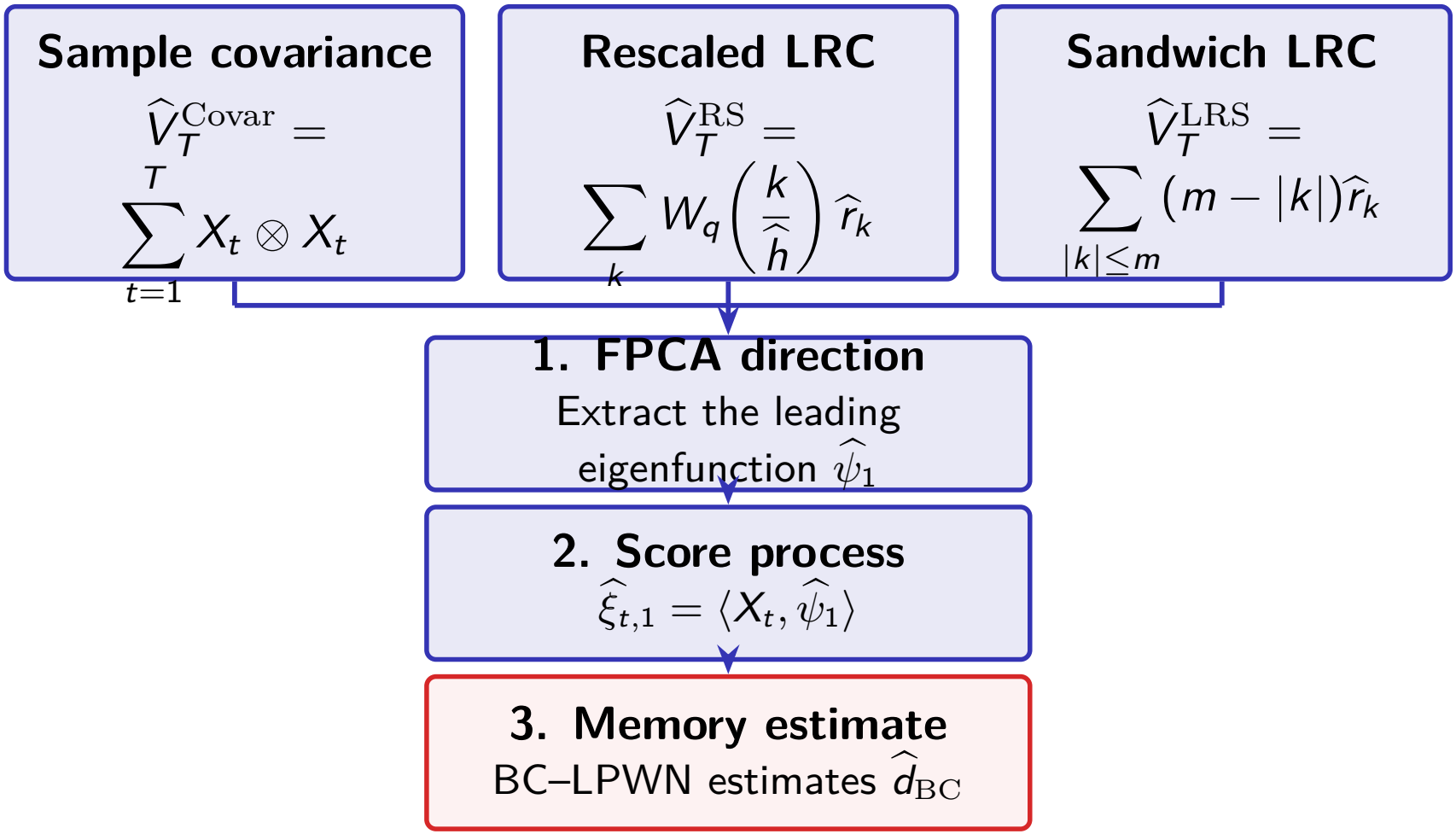
Li et al. (2020, 2023); Kokoszka and Reimherr (2013); Mestre et al. (2021); Klepsch et al. (2017).

This thesis connects the pieces in one estimable workflow



The simulations first isolate each component, then evaluate the complete FARFIMA pipeline.

Step 1: estimate d along the dominant memory direction



Step 2: residual dependence selects the FARMA order

$$\widehat{\Gamma}_h^{(p,q)} = \frac{1}{T} \sum_{t=h+1}^{T} \widehat{\eta}_t^{(p,q)} \otimes \widehat{\eta}_{t-h}^{(p,q)}$$

$\widehat{\eta}_t^{(p,q)}$: residual curve from the fitted FARMA(p, q); $\widehat{\Gamma}_h^{(p,q)}$: its lag-$h$ empirical autocovariance operator; $\|\cdot\|$: Hilbert–Schmidt norm.

Functional Box–Jenkins

Increase complexity; stop at the first residual-white-noise model.

$$V_{T,H}^{(p,q)} = T \sum_{h=1}^{H} \left\| \widehat{\Gamma}_h^{(p,q)} \right\|^2, \qquad H = \lfloor \log T \rfloor$$

Adaptive functional HQ

Compare all candidates; penalise unnecessary FARMA lags.

$$F_T(p,q) = \frac{1}{H} \sum_{h=1}^{H} \left\| \widehat{\Gamma}_h^{(p,q)} \right\|^2$$

$$\mathrm{IC}_T(p,q) = F_T(p,q) + \mathrm{loglog}(T)\, \widehat{\delta}_T(p+q)$$

Step 3: reconstruct hidden shocks before estimating the operators

The moving-average shocks η_t are not observed.

1. Fit a flexible FAR proxy

Use many observed curve lags to approximate the full dependence.

→

2. Recover proxy shocks

Treat the residual curves as estimates of the hidden innovations.

→

3. Estimate jointly

Regress on curve lags and proxy-shock lags to estimate Φ_i and Θ_j.

Hannan–Rissanen turns a hidden-shock problem into a functional regression.

Step 4: the proposed forecast estimates what the benchmark fixes

Benchmark: FARFIMA$(1,1,0)$

$$\Delta X_t = \Phi_1 \Delta X_{t-1} + \eta_t.$$

Fix $d = 1$, $p = 1$, $q = 0$: first-difference the curves and fit FAR(1).

Proposed plug-in forecast

Estimate d, select (p, q), and estimate every operator.

$$\widehat{X}_{T+h|T} = (1-B)^{-\widehat{d}}\widehat{Y}_{T+h|T}.$$

$$\widehat{Y}_{T+h|T} = \sum_{i=1}^{p} \widehat{\Phi}_i \widehat{Y}_{T+h-i|T} + \sum_{j=1}^{q} \widehat{\Theta}_j \widehat{\eta}_{T+h-j|T}, \qquad \widehat{\eta}_{s|T} = 0 \text{ for } s > T.$$

Step 5: one validation set supports two tasks

Training
Fit candidates

Validation
Select + calibrate

Test
Evaluate once

1. Forecast CV chooses a predictive order

Average the h-step squared validation error on the original scale.

$$(\widehat{p}_{\mathrm{CV}}, \widehat{q}_{\mathrm{CV}}) = \arg \min_{(p,q)\in\mathcal{M}} \mathrm{CV}_h(p,q)$$

2. Conformal calibration chooses the band width

Choose the smallest residual multiplier reaching target coverage $1-\alpha$.

$$\widehat{\zeta}_{\alpha,h} = \inf\{z : \mathrm{ECP}_h(z) \geq 1-\alpha\}, \qquad \widehat{X}_{T+h|T}(u) \pm \widehat{\zeta}_{\alpha,h}\gamma_h(u)$$

Evaluation I: structure recovery and point forecasting

Exact order recovery

$$\text{Accuracy} = \frac{1}{R}\sum_{r=1}^{R}\mathbf{1}\{(\widehat{p}_r, \widehat{q}_r) = (p_0, q_0)\}$$

Higher is better

Point forecasting

$$\text{RMSE}_h^{(a)} = \left\{\frac{1}{M_h N}\sum_{m,j} e_{m,h}^{(a)}(u_j)^2\right\}^{1/2}$$

$$\text{RMSE ratio} = \text{RMSE}_h^{(a)} \,/\, \text{RMSE}_h^{(1,1,0)}$$

Below one is better

R: replications; (p_0, q_0): true order; $(\widehat{p}_r, \widehat{q}_r)$: order selected in replication (r).
a: method; h: horizon; $m = 1, \ldots, M_h$: forecast origins; $u_j, j = 1, \ldots, N$: grid points;
$e_{m,h}^{(a)}(u_j) = X_{m+h}(u_j) - \widehat{X}_{m+h|m}^{(a)}(u_j)$.

Evaluation II: interval calibration and sharpness

Empirical coverage probability

$$\mathrm{ECP}_h = \frac{1}{M_h N} \sum_{m=1}^{M_h} \sum_{j=1}^{N} I_{m,h,j}$$

Target: $1 - \alpha$

Interval score: width + miss penalty

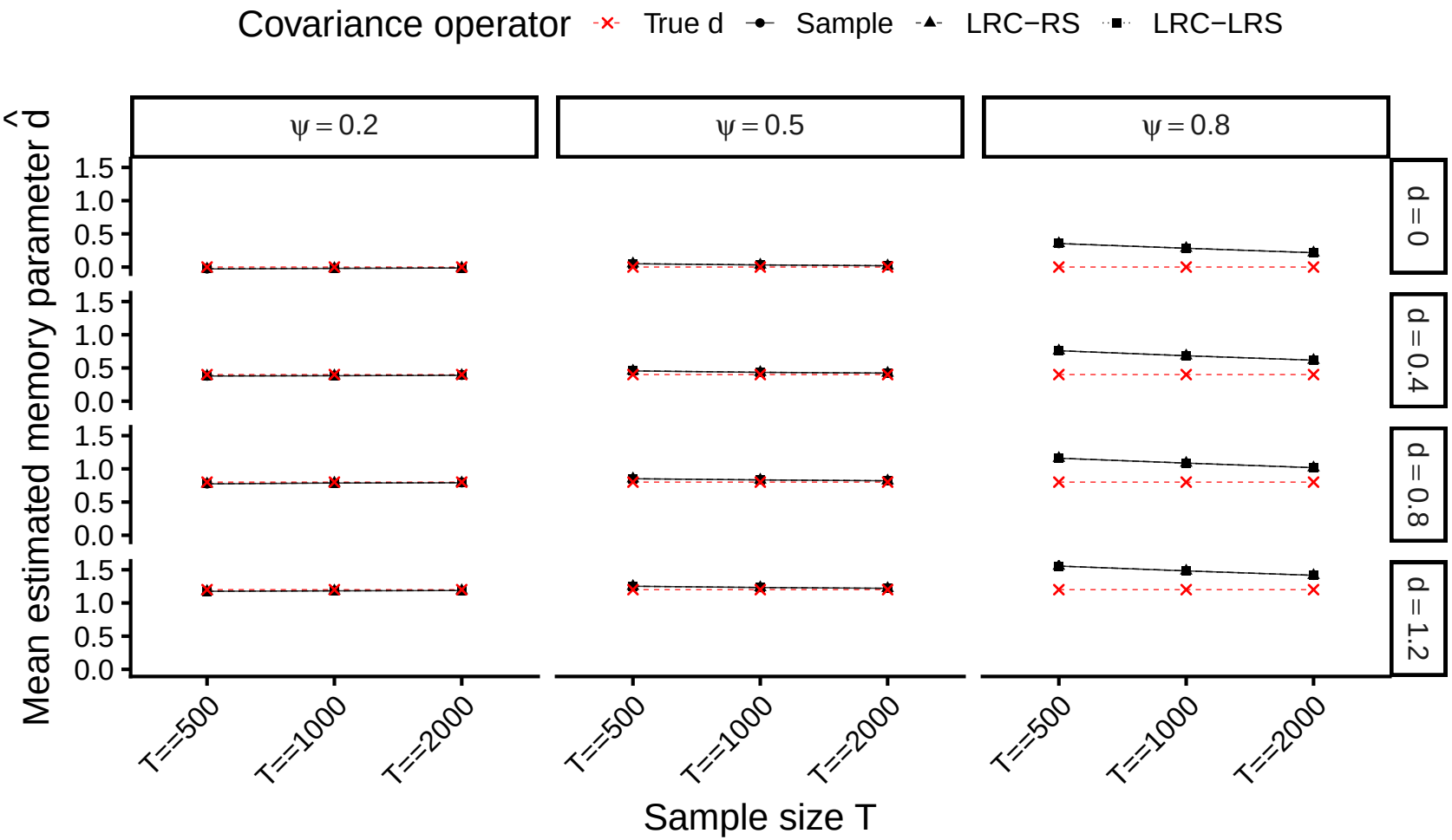
$$\mathrm{IS}_\alpha(X, L, U) = (U-L) + \frac{2}{\alpha}\{(L-X)_+ + (X-U)_+\}$$

$$\text{Score ratio} = \frac{S^{(a)}_{\alpha,h}}{S^{(1,1,0)}_{\alpha,h}}$$

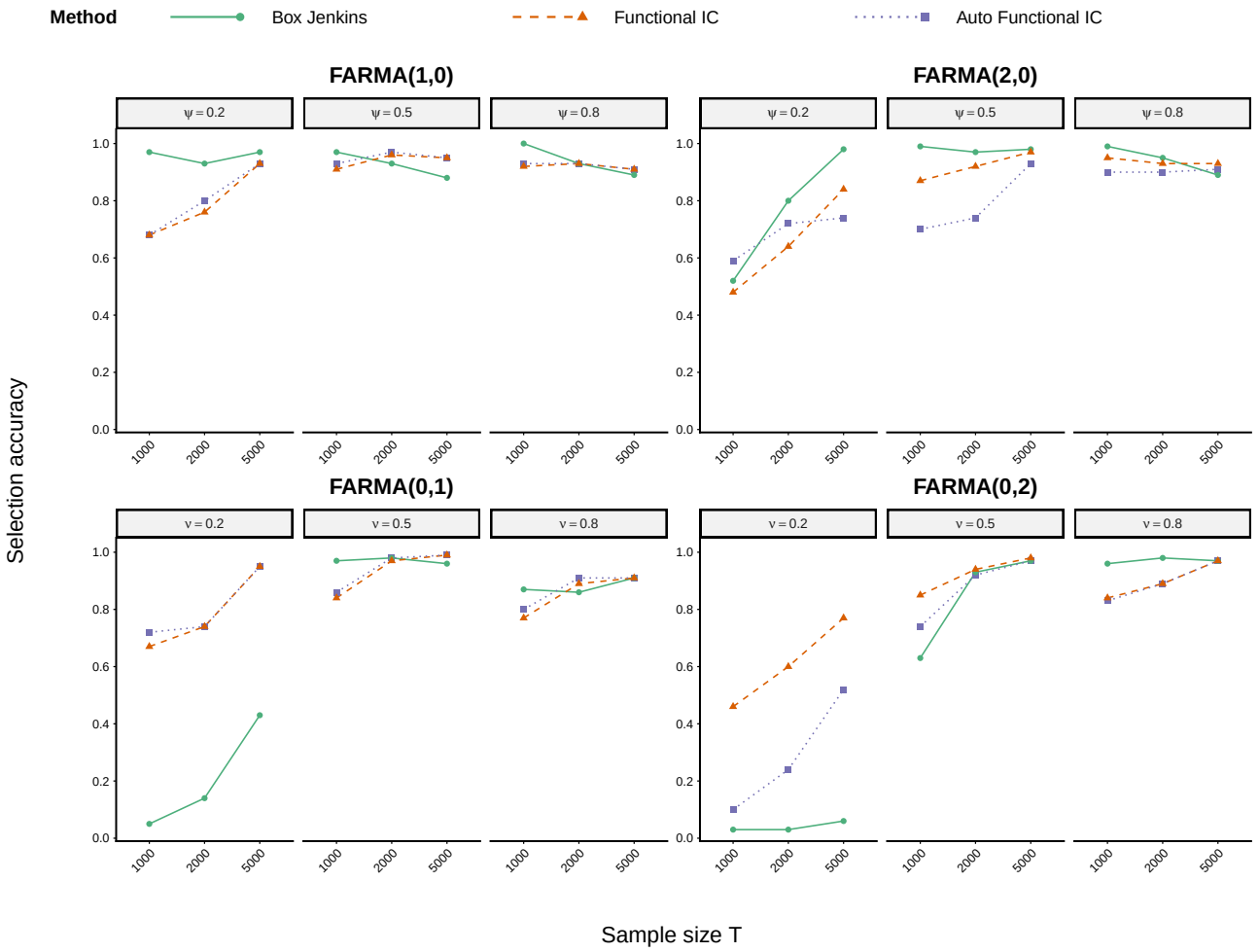
Below one is better

$I_{m,h,j} = \mathbf{1}\{X_{m+h}(u_j) \in [L_{m,h,j}, U_{m,h,j}]\}$.
L, U: band limits; X: realised value; $(x)_+ = \max(x, 0)$; α: miscoverage.
$S^{(a)}_{\alpha,h} = (M_h N)^{-1} \sum_{m=1}^{M_h} \sum_{j=1}^{N} \mathrm{IS}^{(a)}_{\alpha,m,h,j}$: mean interval score across origins and grid points.
m, j, h, a, M_h, N are defined as on the preceding slide.

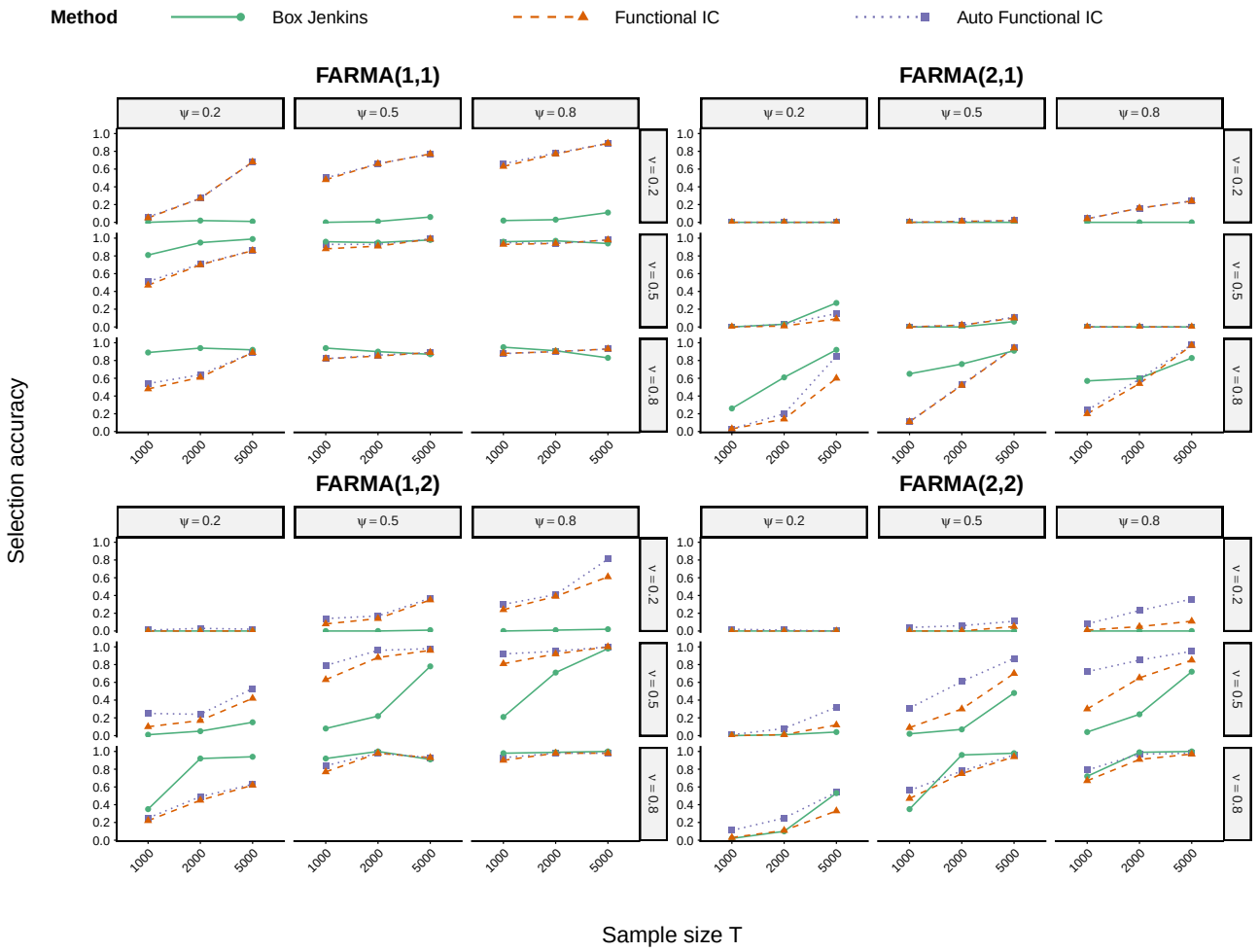
Component 1: covariance operators and memory estimation



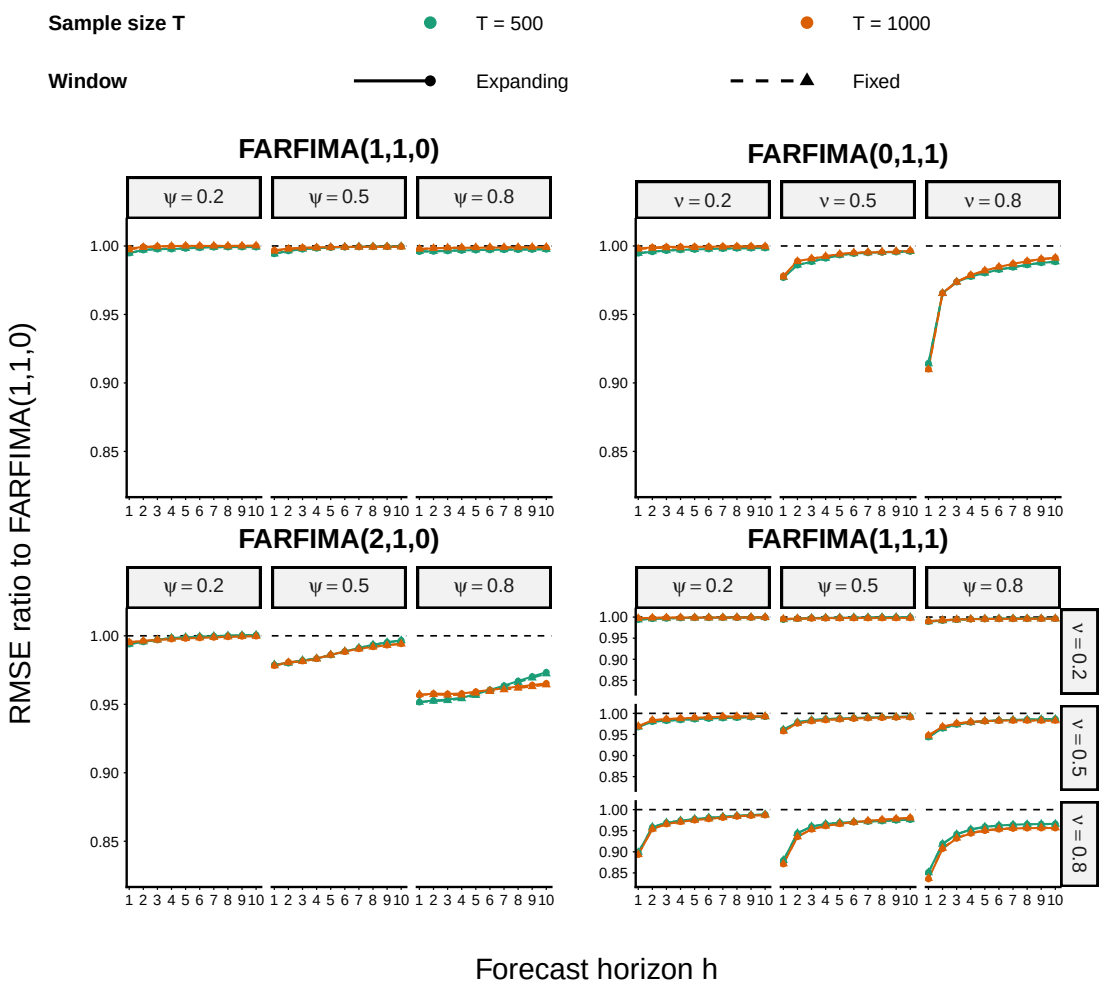
Component 2a: pure FAR and FMA order selection



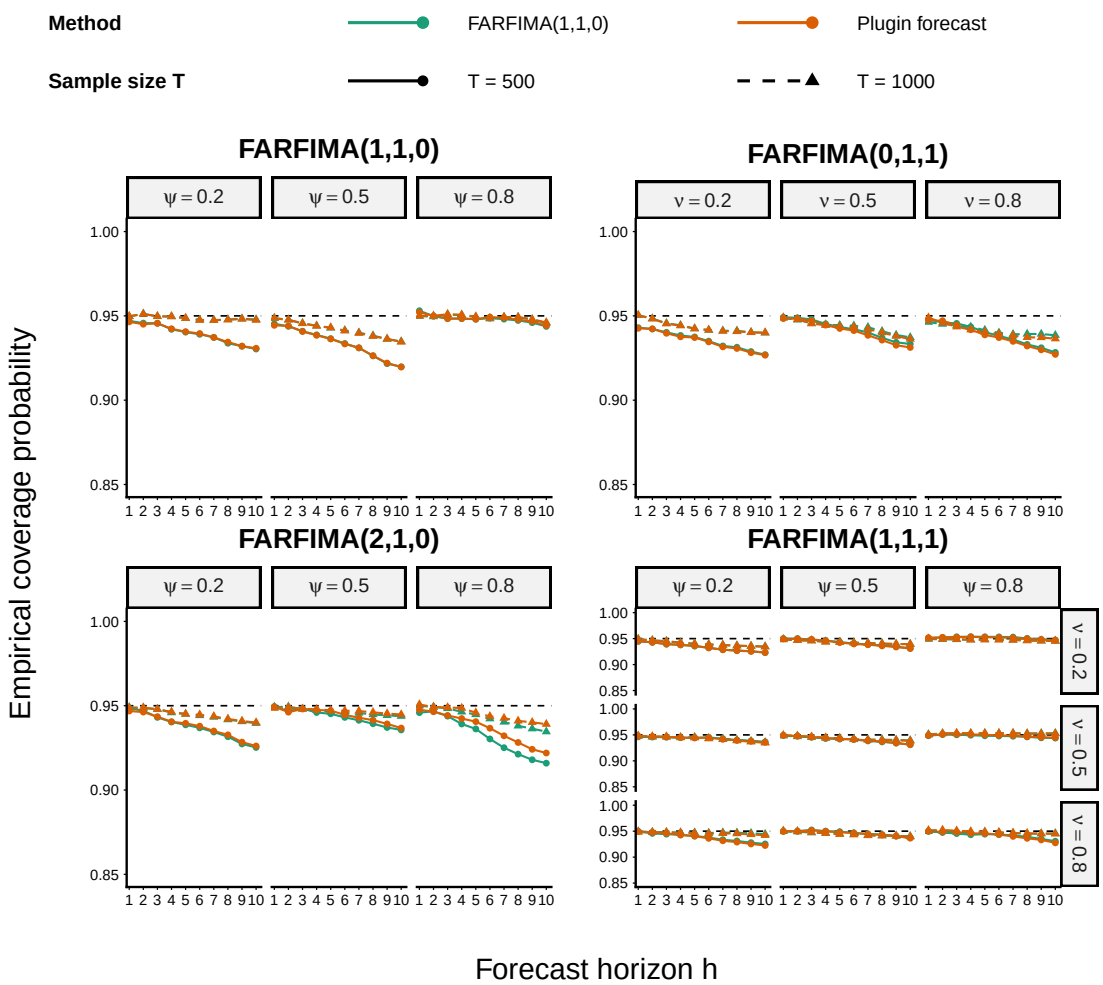
Component 2b: mixed FARMA order selection



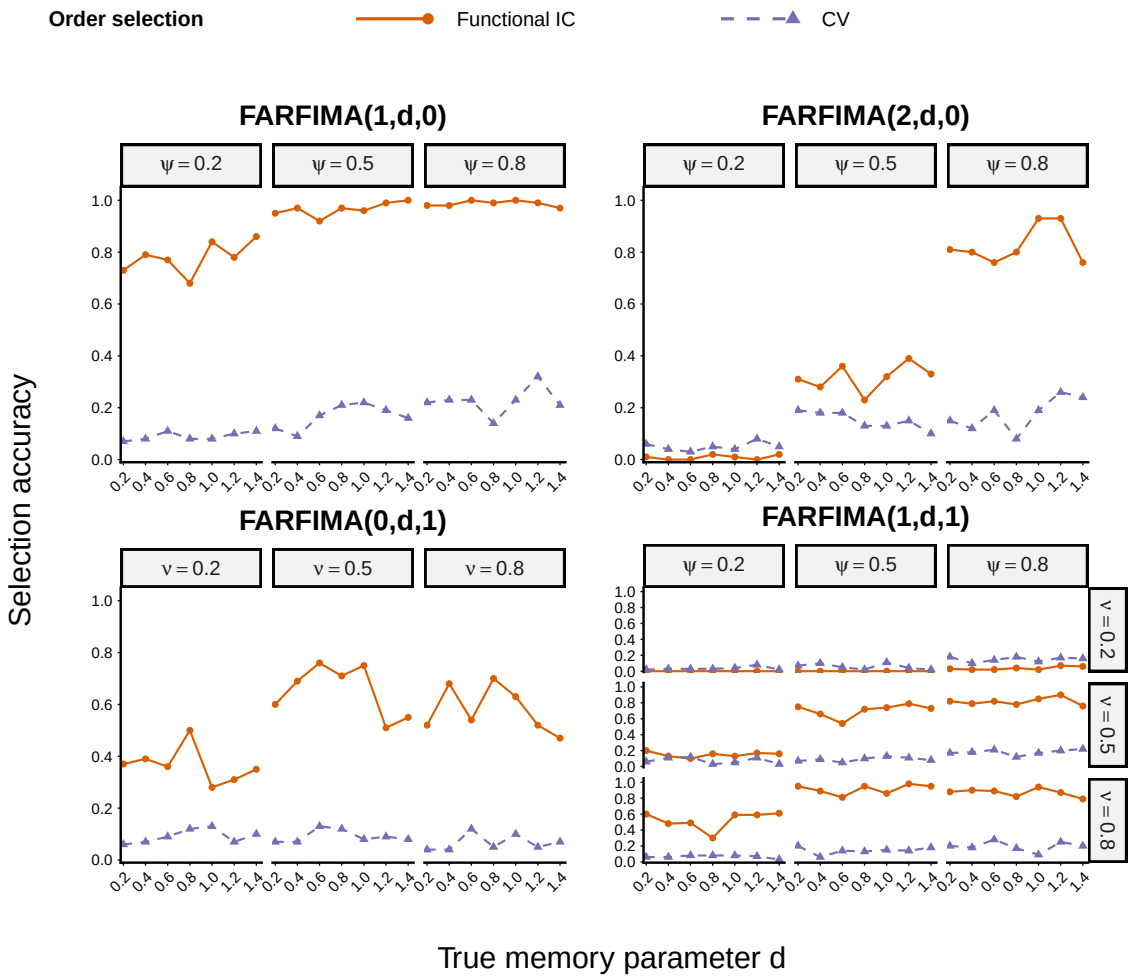
Component 3a: ideal point forecasting with true d and (p, q)



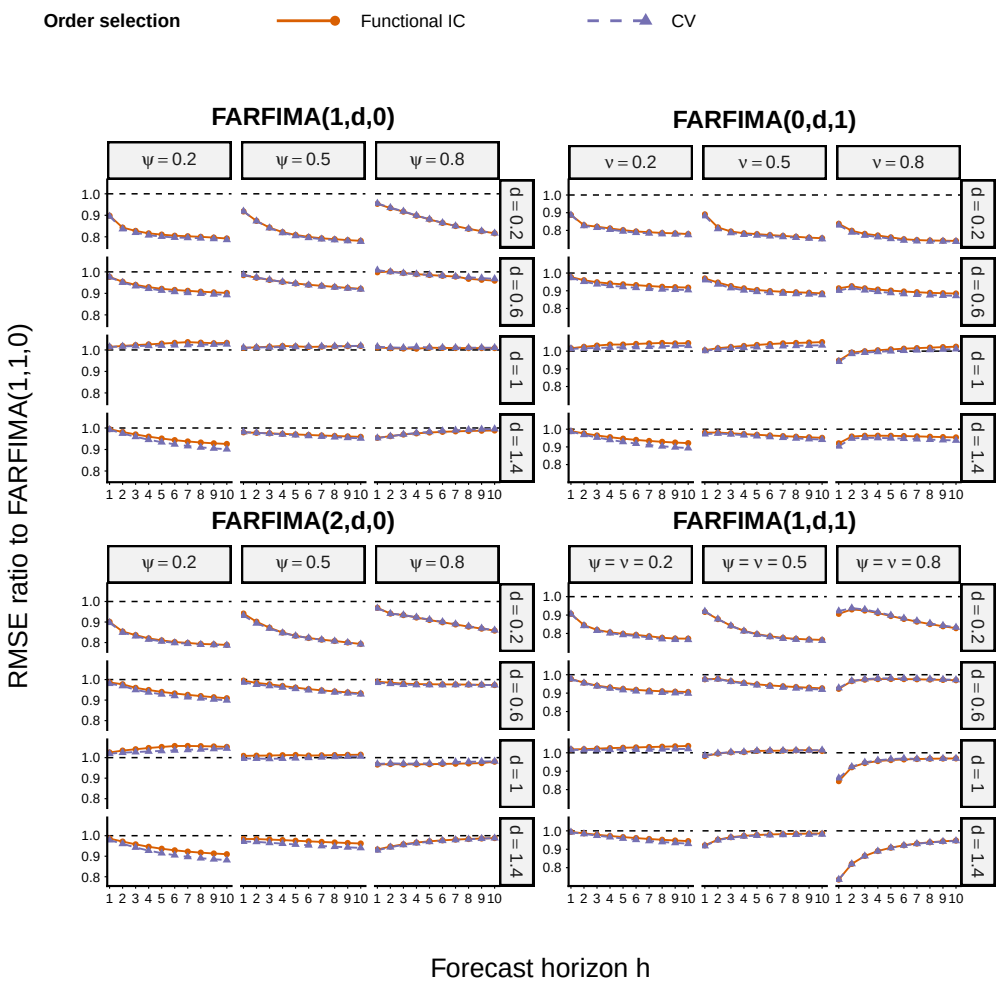
Component 3b: ideal conformal coverage



Unified 1: order selection after estimating d



Unified 2: point forecasting after estimating memory and order



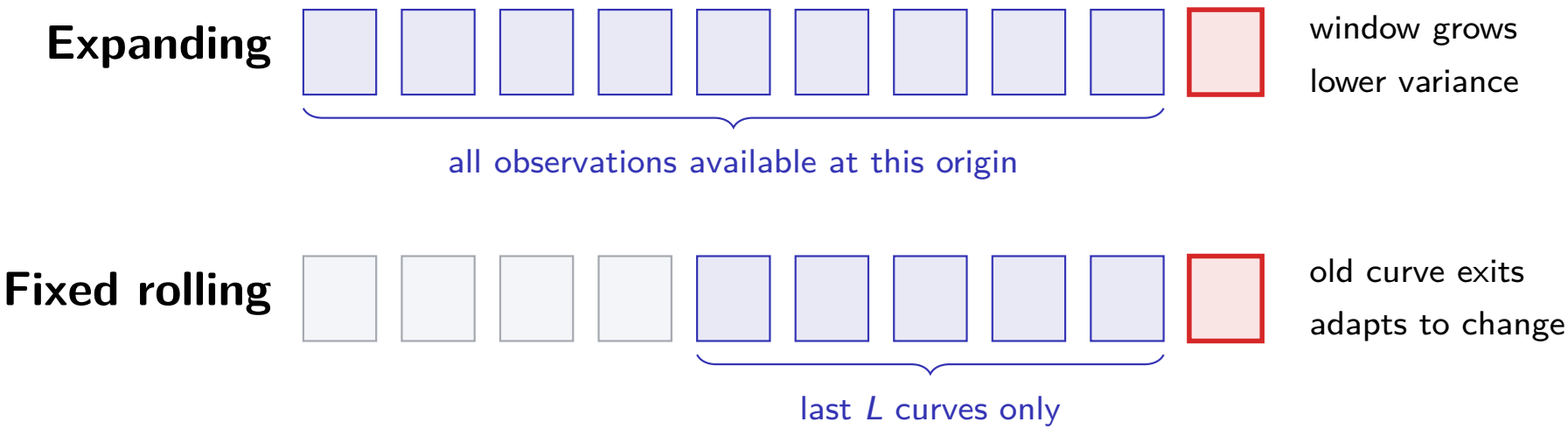
A complete route from persistent curves to calibrated forecasts

1. Separate	2. Select	3. Forecast
Estimate d from the leading score; then difference the curves.	Use residual dependence to select (p, q); adaptive HQ performs best overall.	Estimate FARMA effects, integrate back, and calibrate conformal bands.

Estimate d $\longrightarrow$ select (p, q) with adaptive HQ $\longrightarrow$ forecast and calibrate.

Questions

Backup: expanding versus fixed rolling windows



Expanding windows ask: *does using all available history stabilise estimation?*

Fixed windows ask: *can a recent-history fit adapt when the process changes?*

At every forecast origin, the next h curves remain out of sample; the same rule is repeated through validation and test.

Backup: forecast cross-validation is a predictive order selector

$$\mathrm{CV}_h(p,q) = \frac{1}{|\mathcal{V}_h|N} \sum_{m\in\mathcal{V}_h} \sum_{j=1}^{N} \left\{ X_{m+h}(u_j) - \widehat{X}^{(p,q)}_{m+h|m}(u_j) \right\}^2,$$

$$(\widehat{p}_{\mathrm{CV}}, \widehat{q}_{\mathrm{CV}}) = \arg \min_{(p,q)\in\mathcal{M}} \mathrm{CV}_h(p,q).$$

Adaptive HQ asks a structural question

Which order leaves the most nearly white residuals after penalising complexity?

CV asks a predictive question

Which order forecasts the held-out validation curves most accurately?

CV is therefore a useful byproduct and benchmark, but it need not recover the data-generating order.

Backup: functional Box–Jenkins white-noise hypotheses

$$\Gamma_h^{(p,q)} = \mathbb{E}\left[\eta_t^{(p,q)} \otimes \eta_{t-h}^{(p,q)}\right], \qquad h \geq 1$$

Population lag-h autocovariance operator of the residual process from candidate (p, q).

Null: residual white noise

$$H_0 : \Gamma_h^{(p,q)} = 0, \quad h = 1, \ldots, H$$

Alternative: dependence remains

$$H_1 : \Gamma_h^{(p,q)} \neq 0 \text{ for some } h \leq H$$

$$V_{T,H}^{(p,q)} = T \sum_{h=1}^{H} \|\widehat{\Gamma}_h^{(p,q)}\|^2 \;\Rightarrow\; \sum_{\ell=1}^{\infty} \xi_{\ell,H} Z_\ell^2 \;\approx\; c\chi_\kappa^2$$

$c = \sigma_{V,H}^2/(2\mu_{V,H})$, $\kappa = 2\mu_{V,H}^2/\sigma_{V,H}^2$. Reject for a large statistic; the first candidate not rejected is selected.

Backup: full grid search finds the global IC minimum

$$\mathcal{M} = \{0, \ldots, 5\}^2 \setminus \{(0,0)\}$$

1. Fit every candidate

Estimate the FARMA model for each $(p, q) \in \mathcal{M}$.

2. Compute every IC

Retain candidates with finite $\mathrm{IC}_T(p, q)$.

3. Choose the minimum

Compare all retained candidates once.

$$(\widehat{p}, \widehat{q}) = \arg \min_{(p,q) \in \mathcal{M}:\, \mathrm{IC}_T(p,q) < \infty} \mathrm{IC}_T(p, q)$$

Guaranteed global minimum over $\mathcal{M}$, at the cost of fitting the full grid.

Backup: automatic search follows improving neighbouring orders

1. Estimate the penalty scale

Fit the largest candidate and set $\widehat{\delta}_T = F_T(5,5)$.

2. Select the best starting model

Compare $\mathcal{S} = \{(2,2),(1,0),(0,1)\}$.

3. Search the eight neighbours

Try every valid $(p+\Delta p, q+\Delta q)$, where $(\Delta p, \Delta q) \in \{-1,0,1\}^2 \setminus \{(0,0)\}$.

4. Move only when IC decreases

Continue from the best improving neighbour; stop when none improves the current IC.

Lower computational cost; the returned order is a local, not necessarily global, IC minimum.

Backup: functional IC consistency—criterion and stochastic scale

$$\mathrm{IC}_T(p,q) = F_T(p,q) + \mathrm{loglog}(T)\,\widehat{\delta}_T(p+q), \qquad F_T(p,q) = \frac{1}{H}\sum_{h=1}^{H} \|\widehat{\Gamma}_h^{(p,q)}\|^2$$

$$\delta_T(p,q) = \frac{1}{H}\sum_{h=1}^{H} \|\widehat{\Gamma}_h^{(p,q)} - \Gamma_h^{(p,q)}\|^2, \qquad \widehat{\delta}_T = \min_{(p,q)\in\mathcal{M}} F_T(p,q)$$

Under the draft's uniform projected-operator rate conditions,

$$\widehat{\Gamma}_h - \Gamma_h = O_p(T^{-1/2}) + O_p(r_T), \quad r_T = O_p(T^{-\eta})$$

$$\text{hence } \delta_T = O_p(T^{-1} + T^{-2\eta}), \qquad \mathrm{loglog}(T)\,\widehat{\delta}_T = o_p(1).$$

Claim: for a finite candidate set with identifiable minimal true order (p_0, q_0),

$$\Pr\{(\widehat{p},\widehat{q}) = (p_0,q_0)\} \to 1.$$

Backup: functional IC consistency—underfitting

If (p, q) underfits, at least one population residual autocovariance is non-zero. Therefore $F_T(p, q) - F_T(p_0, q_0) \xrightarrow{p} c_{p,q} := \frac{1}{H} \sum_{h=1}^{H} \|\Gamma_h^{(p,q)}\|^2 > 0.$

The adaptive penalty vanishes on the absolute scale:
$\text{loglog}(T)\, \widehat{\delta}_T \left[(p + q) - (p_0 + q_0)\right] = o_p(1).$

$$\text{IC}_T(p, q) - \text{IC}_T(p_0, q_0) = c_{p,q} + o_p(1) > 0$$

Hence every underfitted candidate is rejected with probability tending to one.

Backup: functional IC consistency—overfitting and conclusion

For an overfitted candidate, both population residual autocovariances vanish, so $F_T(p,q) - F_T(p_0,q_0) = O_p(\delta_T)$.

Let $\Delta k = (p+q) - (p_0+q_0) > 0$. If $\widehat{\delta}_T \asymp_p \delta_T$, then

$$\frac{\mathrm{IC}_T(p,q) - \mathrm{IC}_T(p_0,q_0)}{\delta_T} = O_p(1) + \mathrm{loglog}(T)\frac{\widehat{\delta}_T}{\delta_T}\,\Delta k \xrightarrow{p} +\infty.$$

$$\Pr\{\mathrm{IC}_T(p,q) > \mathrm{IC}_T(p_0,q_0)\} \to 1.$$

A finite union over all wrong candidates combines this result with the underfitting argument and proves exact-order consistency.

The proof uses the finite candidate set, identifiability, uniform covariance-estimation rates, and penalty-scale equivalence stated in the thesis draft.

Backup: functional Hannan–Rissanen algorithm—recover proxy shocks

1. Choose m
Fit a high-order FAR(m), $m > p + q$.

2. Form lags
Stack the last m curves in $Y_t^{(m)}$.

3. Project
Retain empirical eigenspaces.

4. Regress
$\widehat{A}_m = \widehat{\Delta}_m \widehat{\Gamma}_m^{\dagger}$

$$\widehat{\Gamma}_m = \frac{1}{T-m} \sum_{t=m+1}^{T} Y_t^{(m)} \otimes Y_t^{(m)},$$

$$\widehat{\Delta}_m = \frac{1}{T-m} \sum_{t=m+1}^{T} Y_t \otimes Y_t^{(m)},$$

$$\widehat{Z}_{t,m} = Y_t - \widehat{A}_m Y_t^{(m)}.$$

The residual curves $\widehat{Z}_{t,m}$ are proxies for the unobserved innovations η_t.

Backup: Hannan–Rissanen pass 2—joint operator estimates

$$\widehat{\boldsymbol{Y}}_{t,m} = \left(Y_{t-1}, \ldots, Y_{t-p}, \widehat{Z}_{t-1,m}, \ldots, \widehat{Z}_{t-q,m}\right)^{\top},$$

$$\widehat{\Gamma}_{p,q} = \frac{1}{T-m} \sum_{t} \widehat{\boldsymbol{Y}}_{t,m} \otimes \widehat{\boldsymbol{Y}}_{t,m},$$

$$\widehat{\Delta}_{p,q} = \frac{1}{T-m} \sum_{t} Y_t \otimes \widehat{\boldsymbol{Y}}_{t,m},$$

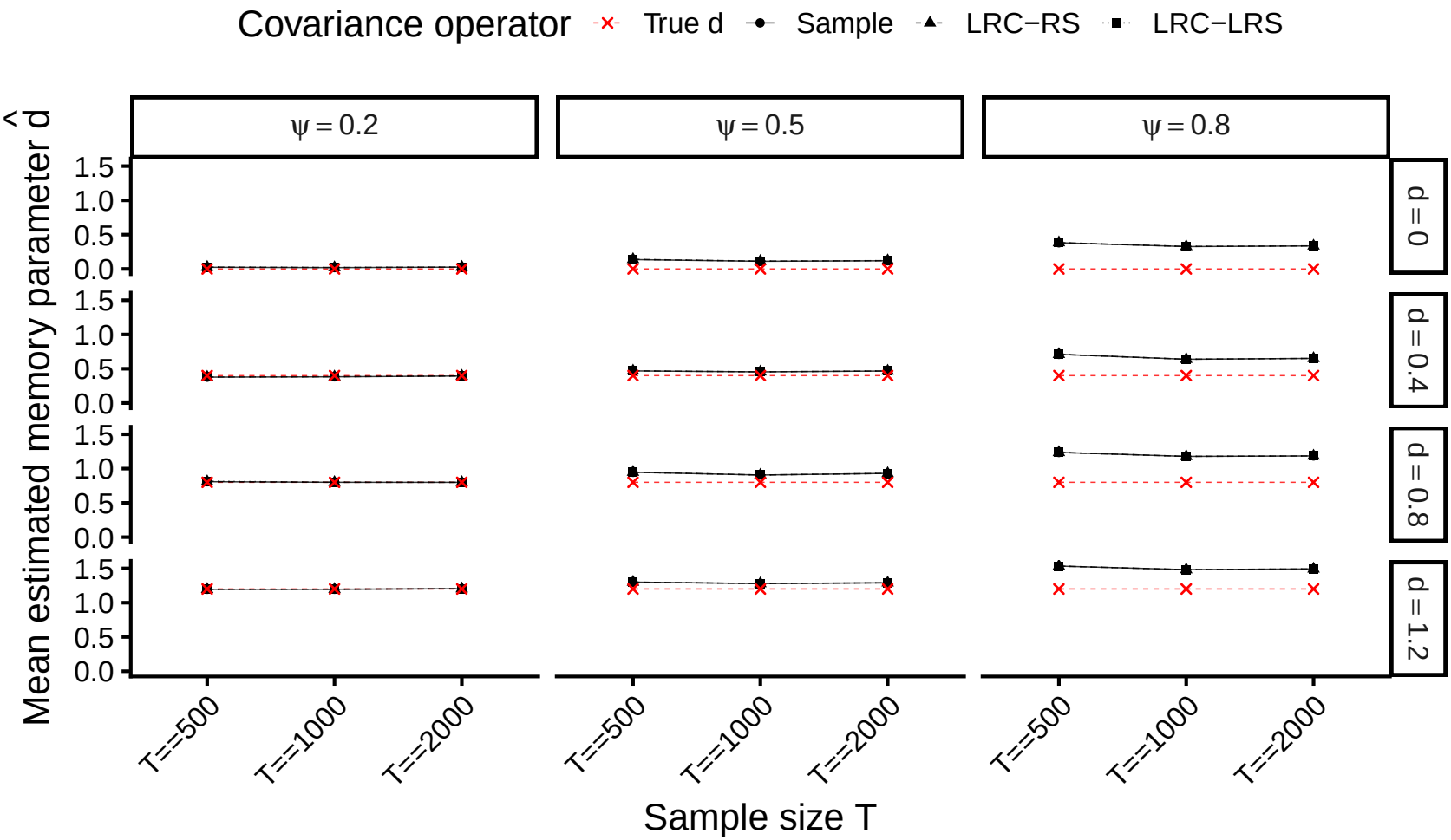
$$\left(\widehat{\Phi}_1, \ldots, \widehat{\Phi}_p, \widehat{\Theta}_1, \ldots, \widehat{\Theta}_q\right) = \widehat{\Delta}_{p,q} \widehat{\Gamma}^{\dagger}_{p,q}.$$

Projection
Makes the infinite-dimensional problem finite

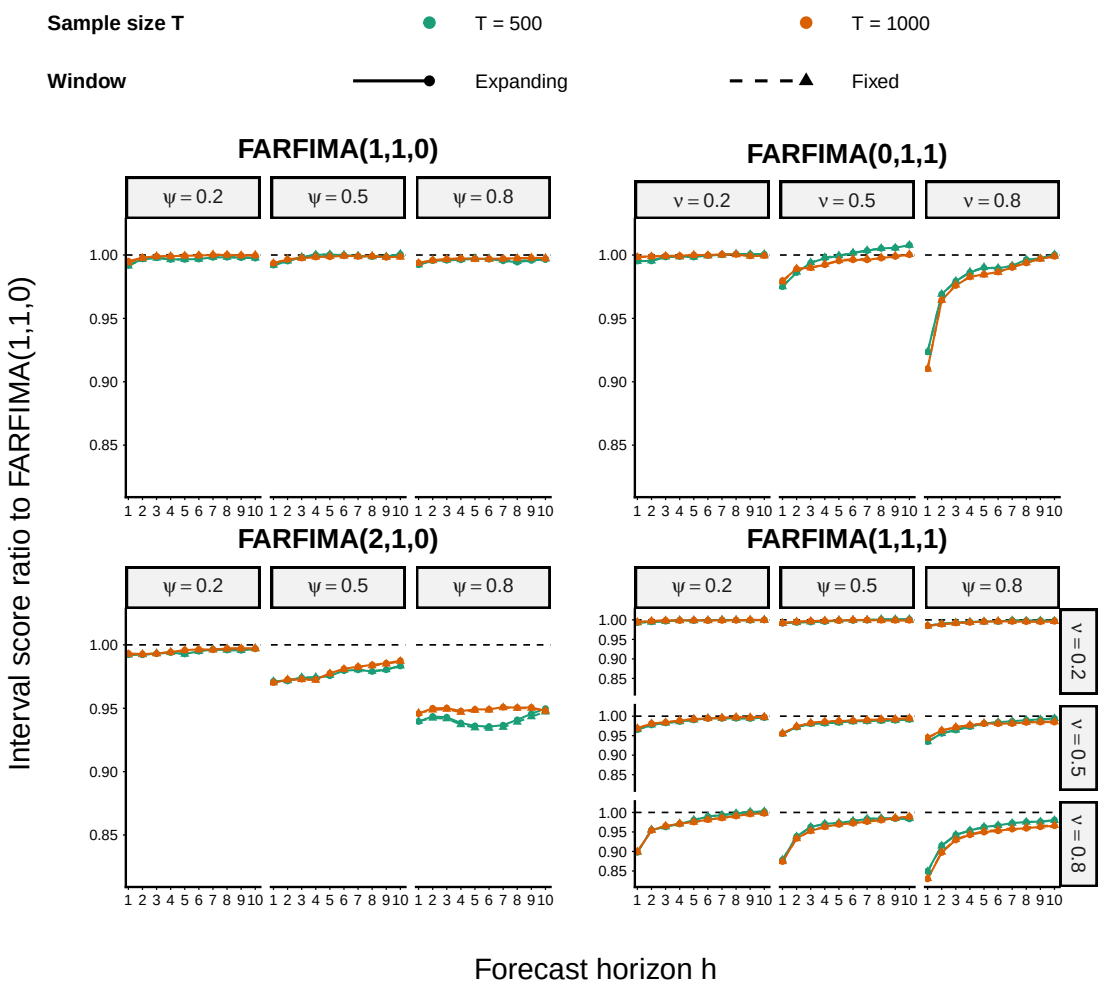
Generalised inverse
Regularises weak directions

Joint regression
Separates feedback from shock propagation

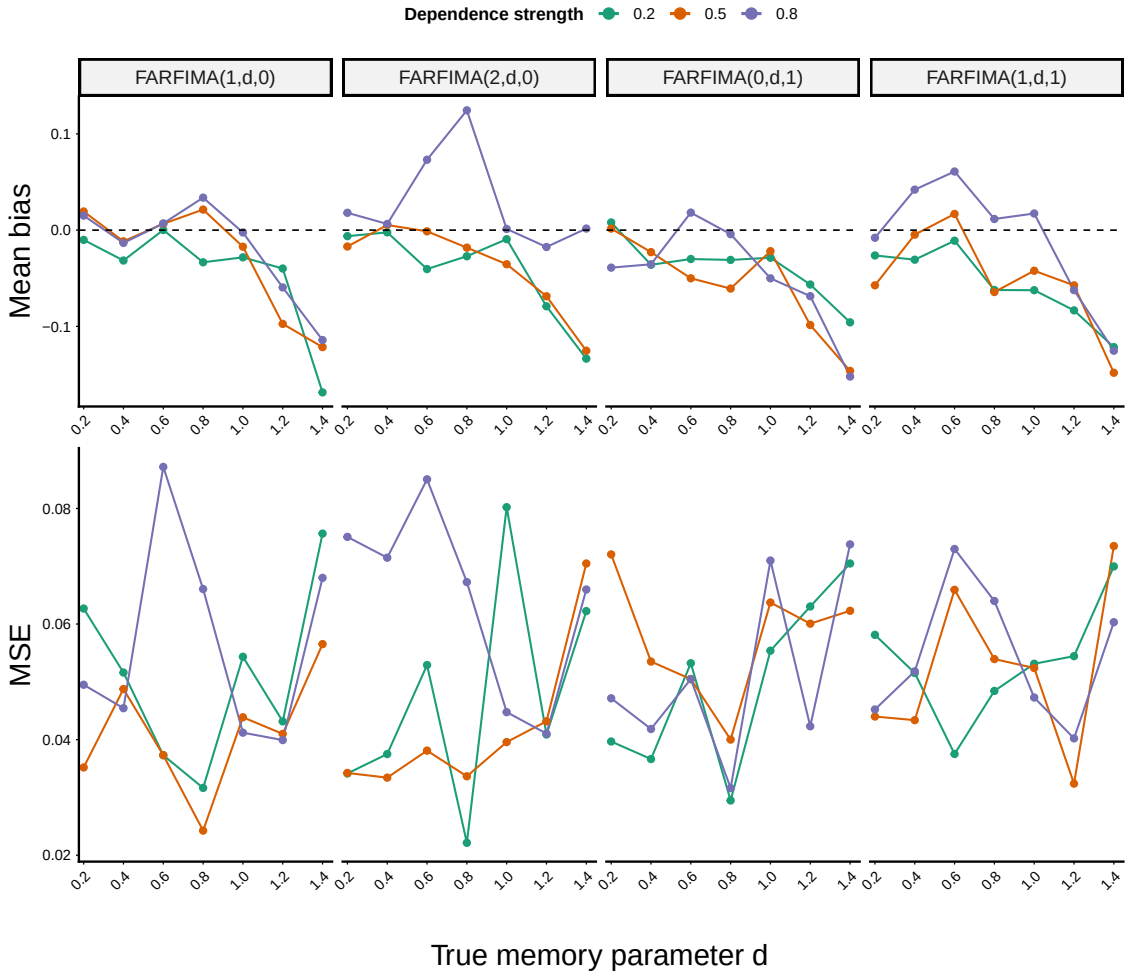
Backup: DFA gives the same covariance-operator conclusion



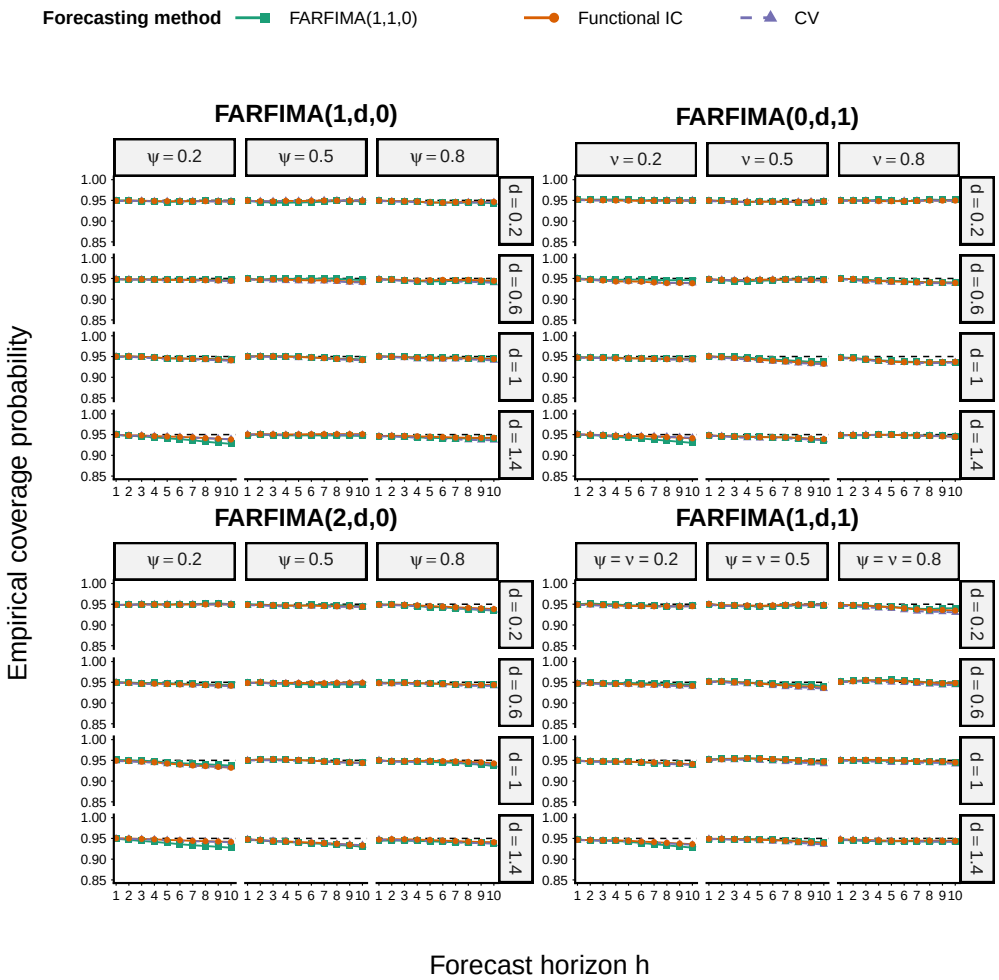
Backup: ideal 95% interval-score ratios



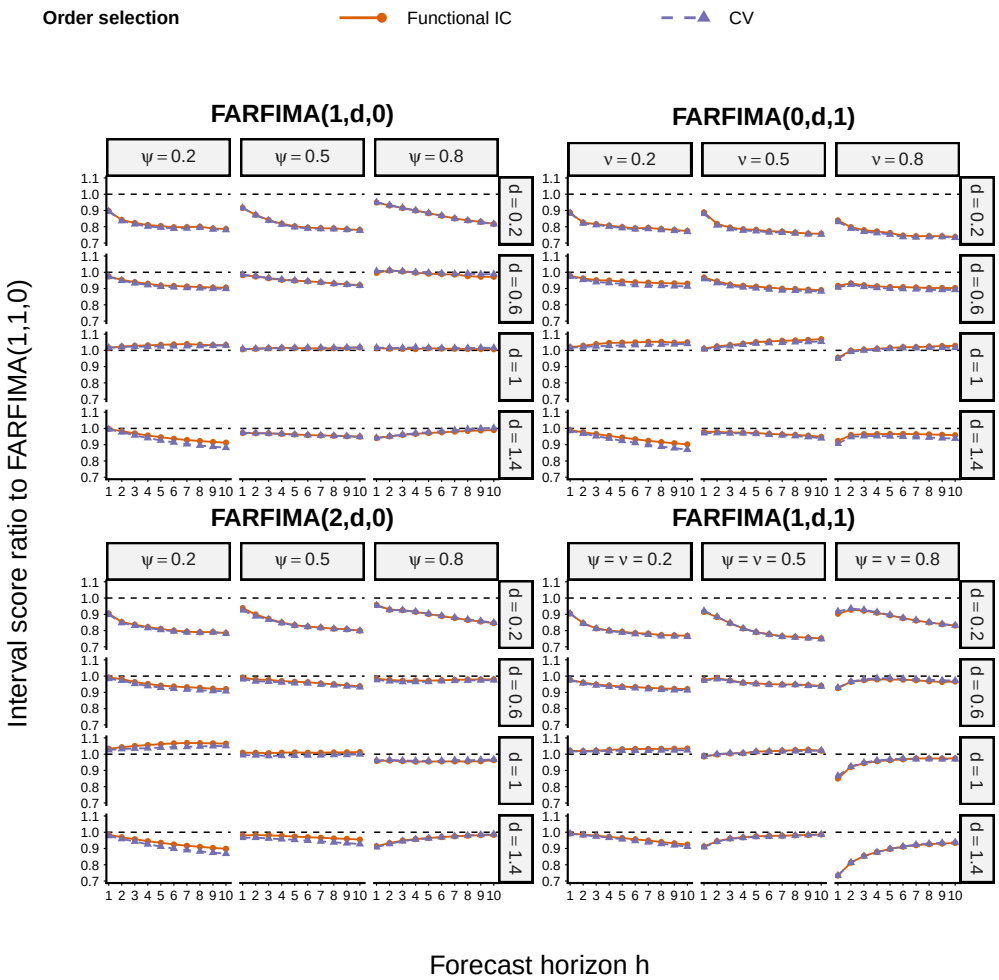
Backup: unified memory estimation



Backup: unified 95% conformal coverage



Backup: unified 95% interval-score ratios



Backup: simulation architecture

Component studies isolate one question at a time

1. covariance operator + memory estimator
2. order selection on stationary FARMA
3. point and interval forecasts with true $d, (p, q)$

Unified study estimates the entire pipeline

Estimate $d \to$ transform $\to$ select $(p, q) \to$ estimate operators $\to$ forecast $\to$ integrate $\to$ calibrate

100 replications horizons $h = 1, \ldots, 10$ expanding and fixed windows